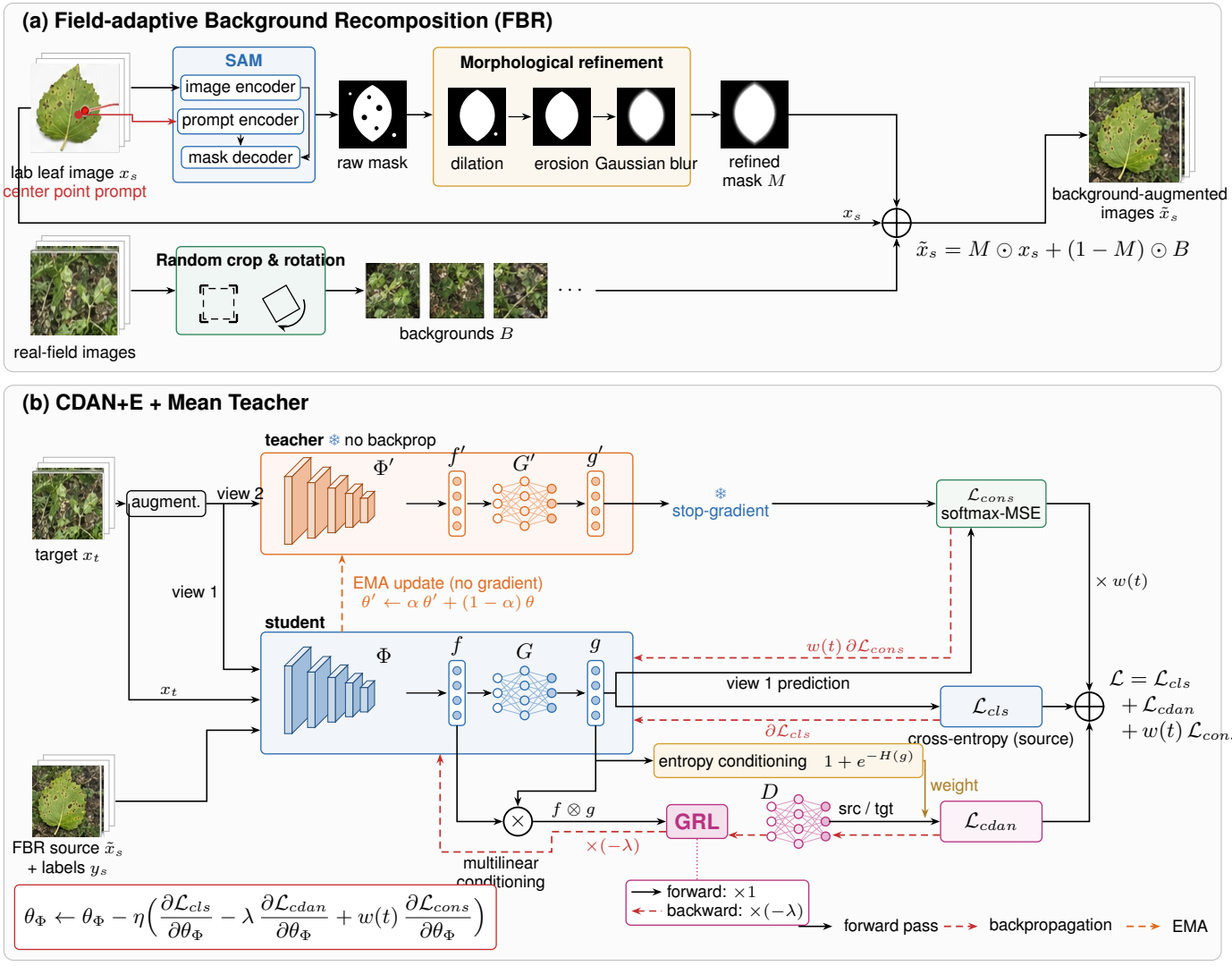

(a) Field-adaptive Background Recomposition (FBR)
SAM
image encoder
prompt encoder
mask decoder
lab leaf image x_s
center point prompt
raw mask
Morphological refinement
dilation
erosion
Gaussian blur
refined mask M
background-augmented images $\tilde{x}_s$
x_s
$\tilde{x}_s = M \odot x_s + (1 - M) \odot B$
Random crop & rotation
real-field images
backgrounds B
(b) CDAN+E + Mean Teacher
teacher ❄ no backprop
Φ'
f'
G'
g'
augment.
view 2
target x_t
view 1
stop-gradient
$\mathcal{L}_{cons}$ softmax-MSE
$\times\, w(t)$
EMA update (no gradient)
$\theta' \leftarrow \alpha\,\theta' + (1-\alpha)\,\theta$
student
Φ
f
G
g
$w(t)\,\partial\mathcal{L}_{cons}$
view 1 prediction
x_t
$\mathcal{L}_{cls}$
cross-entropy (source)
$\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}_{cdan} + w(t)\,\mathcal{L}_{cons}$
$\partial\mathcal{L}_{cls}$
entropy conditioning $1 + e^{-H(g)}$
weight
D
$f \otimes g$
GRL
src / tgt
$\mathcal{L}_{cdan}$
FBR source $\tilde{x}_s$ + labels y_s
$\times(-\lambda)$
multilinear conditioning
$\theta_\Phi \leftarrow \theta_\Phi - \eta\left(\frac{\partial\mathcal{L}_{cls}}{\partial\theta_\Phi} - \lambda\,\frac{\partial\mathcal{L}_{cdan}}{\partial\theta_\Phi} + w(t)\,\frac{\partial\mathcal{L}_{cons}}{\partial\theta_\Phi}\right)$
forward: $\times 1$
backward: $\times(-\lambda)$
forward pass
backpropagation
EMA